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MONEY, CREDIT, AND BANKING LECTURE

Using Market Information in Prudential Bank Supervision: A Review of the U.S. Empirical Evidence

MARK J. FLANNERY

In principle, market and government supervision provide alternative devices for controlling (governing) any type of corporation. Most national governments have instituted nonmarket regulatory mechanisms for banking firms, on the grounds that market-based mechanisms do not adequately discipline banks. But what evidence supports this assessment? Why must governments assure *bank* solvency, but not the solvency of *other* firms? Government oversight naturally displaces private efforts to evaluate and control financial firms. Moreover, if the banking business changes over time—as it assuredly has in the past decade or two—the best combination of government and private supervision may change concurrently. This paper reviews and evaluates the growing empirical literature on private investors' abilities to assess the financial condition of banking firms. The evidence supports the proposition that market investors and analysts could reasonably provide a greater proportion of corporate governance services for large, traded U.S. financial firms.

BANKING FIRMS ARE UNUSUAL (perhaps unique) in the extent of government involvement in their corporate governance. Motivated by a perceived public interest in maintaining financial system stability, governments impose on

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banking firms a broad range of controls and regulations, including restrictions on their permissible activities,¹ corporate structure,² ownership,³ and maximum leverage ratios. This system of prudential regulation coexists with private efforts to assess and control banks' behavior, but little is known about how the two governance systems complement one another. Indeed, empirically comparing two supervisory systems requires careful analysis because market incentives to provide corporate governance services are affected by the extent and quality of government oversight activities.

The ability of market forces to supervise financial firms has become an important policy question as financial firms evolve within increasingly global financial markets. As the pace of financial innovation has accelerated, U.S. regulators have expressed growing interest in the use of market-related oversight to supplement their own methods of analysis, especially for large, complex banking organizations.⁴ While a more market-oriented approach to prudential regulation has considerable appeal on theoretical grounds, many regulators bring to this issue a substantial suspicion about the usefulness of market prices and market assessments. This paper reviews empirical evidence about market participants' abilities to identify and control risk in banking firms.⁵ I concentrate on three specific aspects of market information about bank activities.

1. Are market valuations of banking firms *accurate*?
2. Do investors identify important changes in bank condition or risk exposure more *slowly* than government supervisors?
3. Are market valuations of banking firms irrationally *contagious*? In other words, do investors infer inappropriate things about one firm from another firm's experience or condition?

The available empirical evidence suggests that regulators could expand their reliance on market discipline, at least for large, publicly owned institutions.

Any supervisory (governance) system must both *assess* bank condition and promptly *implement* disciplinary actions when they are required. Numerous studies have evaluated whether markets accurately assess bank condition. A much more limited (though growing) amount of literature compares the timeliness of private versus governmental assessments of bank condition. However, I have identified few studies

1. The United States has most severely limited the investment banking activities of commercial banks. U.S. banks are also entirely excluded from participating in nonfinancial commerce.

2. U.S. banks may undertake many activities only through separate holding company affiliates, unlike the "universal bank" form of organization that is common in Europe. The Federal Reserve has long insisted that a BHC should serve as a "source of strength" for its subsidiary banks, and FIRREA formalized this requirement. With private corporate governance, individual firms could negotiate whether or not to eschew this limited liability protection for their banking subsidiaries.

3. U.S. ownership restrictions preclude a firm from controlling both financial and commercial businesses. The market for corporate control in banking is consequently weaker than in other sectors of the economy (Prowse 1997; Gorton and Rosen 1995; Hubbard and Pallia 1995).

4. Increased reliance on private assessments has been particularly important in the design of U.S. capital standards for market risk exposure. These standards use a bank's own internal models to measure its risk exposure, on the grounds that a regulator cannot assess risk more accurately than the bankers themselves do.

5. Gilbert (1990) and Berger (1991) evaluated the earlier parts of this literature, but many important institutional and regulatory changes have occurred since those reviews were undertaken.

comparing the market's versus the government's ability to discipline (that is, remedy) problem situations that develop at specific banking firms. The confidentiality that surrounds U.S. supervisory arrangements has not permitted careful study of this question—though it surely requires empirical analysis. Although I briefly discuss private versus public effectiveness in imposing disciplinary pressure, most of the paper concerns quality assessments and the flow of capital market information.

Survey papers necessarily treat numerous issues. Some readers may wish to focus on a subset of the topics addressed, and I have tried to make each section of the paper relatively self-contained. Section 1 sets the stage for the ensuing literature review, by discussing the purpose of a corporate governance structure and describing key features of the existing public-private mechanism for controlling banking firms. Section 2 summarizes the types of market information that are available for U.S. banks, and concludes that these information sources can provide insights about a large proportion of the banking system. Section 3 examines the impact of a bank's condition on the cost and composition of its liabilities. Although bank liabilities seem like the most natural place to seek evidence about potentially contagious runs, an extensive federal safety net has protected many bank liability holders since 1933. Recent empirical evidence about liability rates and flows may therefore underestimate the tendency of banks to suffer contagious runs. Section 4 therefore evaluates whether equity market studies (for both financial and nonfinancial firms) provide evidence that outside investors make "contagious" inferences about firm conditions. Section 5 reviews the pre-1933 U.S. experience, to assess whether market forces provided reasonable stability in the days before a government safety net had been established. Section 6 turns from the *accuracy* of market assessments to consider their *timeliness*. Does supervisors' access to confidential information permit earlier identification of material changes in bank condition? The (limited) available evidence provides no support for the notion that market or government assessments are uniformly more timely. The limited evidence about how market and government agents *control* bank actions is summarized in section 7, whose main conclusion is that we need further research on this important question.

Having surveyed and evaluated the empirical evidence, Section 8 discusses the potential means of incorporating market information into the process of bank regulation. I suggest here that market information may improve two features of the oversight process: it may permit regulators to identify developing problems more promptly, and it can provide regulators with the incentive and justification to take action more quickly once problems have been identified. I have compiled a list of policy-related questions requiring further research. The paper concludes with a summary of the review's main points and their policy implications.

1. ALTERNATIVE CORPORATE GOVERNANCE ARRANGEMENTS

A recent survey paper begins with a definition: "Corporate governance deals with the ways in which suppliers of finance to corporations assure themselves of getting a

return on their investment” (Schleifer and Vishny 1997, p. 737). In broad terms, these “assurances” affect two dimensions of firm behavior: the organization of its “initial” activities into a specific corporate form, and the conditions under which that initial form will be revised in response to subsequent developments. A firm’s governance structure addresses agency costs and divergent incentives among its shareholders, managers, and fixed income (that is, bond) claimants. Firms select the range of their real-sector activities, and how those activities are organized within a set of corporate entities. Organizational form can importantly affect the riskiness of the claims a firm sells to raise external funds.⁶

A firm needing funding can approach the market with a variety of securities offerings. Investors infer the likely cash flows to various securities being offered in the market, and purchase those which seem to offer the best risk-return trade-offs. Investors’ risks depend on both the security’s features and the firm’s corporate structure: secured versus unsecured debt, whether a new activity is undertaken in a separate subsidiary, the capitalization of that subsidiary, cross-guarantees among members of the corporate family, etc.⁷ If a firm finds its securities bring a low price, it may alter its activities or organizational form in response. *Ceteris paribus*, fund-raisers will select corporate structures which minimize their investors’ risk concerns, subject to internal constraints. Corporate governance is particularly important for highly levered firms, whose shareholders (who presumably control management) have interests that diverge from those of bondholders. Levered firms frequently accede to restrictive debt covenants which govern their permissible activities, corporate structure, and the allocation of cash flows to particular claimants. By including restrictive covenants in their debt obligations, firm owners can obtain cheaper or more available debt funding (Smith and Warner 1979). Because they recognize the riskiness of proposed activities *ex ante*, bond covenants force owners to consider the full range of possible outcomes from their operations.

Corporate governance is a dynamic process which requires a firm’s stakeholders to monitor its performance continuously. A corporate structure which is appropriate when equity equals 50 percent of total assets may impose deadweight costs on the firm if losses reduce its equity to 30 percent. The firm will then find that it cannot borrow additional funds, or that the additional funds are very costly, and will consider revising its corporate structure and/or real activities in response to this market pressure. Firms that have suffered large losses may also be taken over, or their boards may replace top managers (Mikkelsen and Partch 1997).

An important component of market discipline’s power is that private investors can

6. A firm with two lines of business, for example, could operate as an integrated company or as a holding company with two self-financing subsidiaries (Flannery, Houston, and Venkataraman 1993). Even a firm with only one line of business can select alternative means of production, which differ in the extent of their operating leverage and their break-even volumes.

7. For example, in 1992 several prominent U.S. banks and investment banks responded to market concerns about derivatives counterparty risk by establishing separately capitalized, AAA-rated subsidiaries for derivatives trading. This change in corporate organization reduced the credit-related costs of dealing with others. Similar structural issues of current policy concern include the combination of banking and commerce, and whether banking firms should engage in “nonbank” activities within the bank or as a separate holding company subsidiary.

specialize in evaluating certain types of securities (firms). Some investors prefer high-quality, low-risk investments, while others have a comparative advantage in evaluating risky situations. Firms with special risks or needing special types of corporate governance can deal with investors who believe that they can monitor and control complex firms particularly well. This self-selection permits economies of specialization in the market for corporate governance. In equilibrium, different types of firms operate with different financial and operational structures (Demsetz 1983; Demsetz and Lehn 1985). By contrast, the government safety net provided to banks is generally accompanied by “one-size-fits-all” prudential regulations that constrain banking firms’ activities and corporate organizational form.

Government prudential regulation of financial firms is broadly motivated by a concern for maintaining stability in the financial system. The need for *government* oversight to assure stability in this sector (when it is not required in most other sectors) is motivated by two broad considerations. First, much government regulation reflects the belief that bank depositors cannot effectively protect themselves.⁸ Dewatripont and Tirole (1994) argue that numerous, small depositors cannot effectively identify or control bank risks because of information costs and coordination problems. In their view, government remedies a market failure by acting in place of the atomistic depositors.⁹ A second basis for government prudential supervision derives from the nature of bank assets, which are alleged to be unusually informationally opaque (and therefore illiquid):

Bank loans are customized, privately negotiated agreements that, despite increases in availability of price information and in trading activity, still quite often lack transparency and liquidity. *This unquestionably makes the risks of many bank loans rather difficult to quantify and to manage.* (Greenspan 1996, pp. 1–2, emphasis added)

Government supervisors may understand a bank’s condition more precisely because of (some) unique access to private information. More importantly, the combination of opaque assets and short-term deposit liabilities is said to make banking firms potentially unstable (Goodhart 1988; Gorton 1988; Dwyer and Gilbert 1989).¹⁰ Not only may an individual bank’s solvency be threatened by liquidity problems, but financial problems at one bank may cause investors to suspect similar problems at other banks, causing contagious runs:

It is probably fair to say that there is considerable agreement among central bankers and other economic policymakers that [banks’] unique balance sheet structure creates an inherent *potential instability in the banking system*. Rumors concerning an individual

8. The fact that banks are not subject to SEC disclosure requirements when they issue deposit obligations suggests that the government chartering and oversight process assures these “securities” safe.

9. Of course, we should not take the structure of banking firms—with numerous small depositors—as exogenous. Market forces could conceivably design securities appropriate for uninformed small investors (Gorton and Pennacchi 1990), and “narrow” banks would address the same concern via government mandate (Litan 1987; Pierce 1991). If markets failed to provide sufficient investment alternatives, it has long been recognized that the government could issue small-denomination Treasury securities as a safe repository for small savers’ wealth.

10. Some commentators view the combination of short-term deposits and illiquid assets as an aberration. Flannery (1994) argues that short-term debt may be the optimal way to minimize contracting costs for a firm engaged in opaque lending.

bank's financial condition—even if ill-founded—can spark a run by depositors and other creditors . . . Especially if the distressed institution is large or prominent, the panic can spread to other banks, *with potentially debilitating consequences for the economy as a whole.* (Kelley 1997, emphasis added)

Special government treatment for banking firms may therefore be viewed as correcting a market externality based on imperfect information about the true value of opaque asset portfolios.

Concerns about opaque assets have generated the two main features of government safety nets: the lender of last resort and deposit insurance, both of which make the government a contingent claimant on the banking system's assets.¹¹ As such, regulators monitor and control banks' risk exposures in order to limit the taxpayer's liability for banking losses. The government's prudential activities importantly affect private incentives to evaluate and discipline banking firms. Two examples of this substitution are provided by Prowse (1997) and Goyal (1997). Prowse argues that bank ownership restrictions inhibit the market's ability to discipline poorly performing banks, and finds that federal regulators provide the most common type of external discipline for banking firms. Goyal's (1997) example illustrates the market's interest in replacing lost government oversight. He investigates the use of restrictive covenants in bank subordinated debentures, and finds that their incidence increased after federal legislation (in the early 1980s) eased regulatory restrictions on permissible bank assets. An increase in the extent of government oversight induces the market to provide less oversight, while the government's withdrawal from oversight and risk bearing will surely elicit greater activity by market investors.¹²

Table 1 provides a framework for comparing the efficacy of government versus market systems of corporate governance. Not surprisingly, each type of supervision has some comparative advantages, making it difficult to choose one system over the other on the basis of theory alone. Government agents probably have a cost advantage over multiple private analysts, and their access to inside information about firm condition may be superior for two reasons: examiners can force managers to reveal information, and a single government agency does not suffer the coordination (free-rider) problems associated with having many fragmented stakeholders. These government advantages may be partially offset by governmental constraints which make it difficult to pay competitive market wages. In addition, regulators' incentives to identify and remedy problems promptly have been questioned, particularly following the thrift debacle (Kane 1997). Potentially substantial agency problems also exist in the market sector. For example, private auditors may be reluctant to disseminate "bad" information about client firms.¹³

11. A third important component of the safety net is Fed-provided payments finality on its large-dollar transfer system, which is said to protect the nation from systemic risk. This element of the federal safety net blunts bank incentives to control their own risks: if bankers believe that their large counterparties will be protected from failure, they have scant incentive to diversify bilateral credit exposures. The extent of these risk exposures may then imply (to some) that systemic risk is an inherent part in the payments system.

12. Fishman and Hagerty (1992) make a similar point when examining the effect of insider trading. The intuition underlying their model is that outside investors may stop seeking information about a firm if its insiders will always trade on important information before an outsider can possibly learn about it.

13. The SEC recently expressed concern "that the accounting industry's push into consulting and other

TABLE 1
COMPARING MARKET AND GOVERNMENT SUPERVISORY SYSTEMS

	Government Supervisors	Market Analysts and Investors
ASSESSING BANK CONDITION		
Cost	Cheaper to have one monitor than many.	Free-riders.
Access	On-site examinations and ability to compel revelation of information.	Cannot compel information release; firm reluctant to disclose "trade secrets", etc.
Analytical Skills	Hiring subject to government rules; standardized form of analysis may limit flexibility.	Market wages attract people with greatest abilities; different types of analysis may be undertaken across investors.
Incentives	Government supervisors may confront less pressure to identify or rectify problem situations.	Firms and investors have their own funds (reputation) at stake; tendency to free-ride when there are many stakeholders.
APPLYING DISCIPLINE		
Ability to Act on Information	Extensive "fairness" and due process constraints impede rapid action.	Dispersed ownership; free-riders.

When it comes to disciplining private firms, government agents operate under rigid procedural constraints designed to insure that all regulated institutions are treated "fairly."¹⁴ These procedures often slow the imposition of corrective measures, even when problems have been accurately identified (O'Keefe and Reidhill 1997). By contrast, the market is not expected to be "fair"—at least not to the same extent. Market disciplinary forces can therefore take effect rapidly, though only if the government safety net does not mute market incentives to act quickly when a problem is first recognized.

Although the federal safety net began as a relatively limited endeavor (Flood 1992), it expanded during the 1980s to provide effective protection to all liability holders of (at least) large financial firms. The Comptroller's "too big to fail" (TBTF) statement in September 1984 was the most explicit example of this expanded protection.¹⁵ Congress limited the insurance agencies' discretion in the Federal Deposit Insurance Corporation Improvement Act (FDICIA), which became law on December 19, 1991. FDICIA mandated "least-cost resolution" of failed institutions, except in cases where imposing credit losses on uninsured claimants would endanger the financial system's stability. Recent bank failures have indeed been more likely to impose losses on uninsured depositors (Helfer 1997), but there remains the strong possibility that a promi-

lines of business could compromise their auditing work for public companies" (Schroeder and MacDonald 1997, p. B10). Flannery and Houston (1998) and Dahl, Hanweck, and O'Keefe (1995) provide evidence about the relative abilities of auditors and government examiners to elicit disclosure of material financial information.

14. For example, the Fifth Amendment prohibits "taking" of private property without "due process."

15. See Carrington (1984) and O'Hara and Shaw (1990). White (1990, especially ch. 8) described TBTF developments in the thrift sector.

nent bank's failure will be met with blanket guarantees designed to maintain financial sector stability.¹⁶

The presence of government oversight and credit guarantees complicates the process of inferring market assessments from historical data. If *de facto* and conjectural government guarantees blunt investors' risk exposures, clear evidence of market discipline may be difficult to find. Finding that bank risks are not accurately imputed into liability rates may not mean that investors cannot understand true bank condition, but only that they believe the government will insulate them from losses. Finding even a weak connection between bank condition and liability prices would suggest that investors are pricing bank-specific default probabilities. Conversely, government oversight may *enhance* the market's ability to understand and value bank condition, for example, if on-site examinations or government-imposed reporting requirements cause the dissemination of valuable information.

Market and government oversight can be combined in (at least) two ways. First, government could retain its present responsibility for the solvency of banking firms, but formally incorporate market information into the process of identifying and correcting problems at the largest banks. This change represents only a qualitative shift in the traditional activities of government regulators. A more radical reform would have government withdraw from the direct regulation of banks that are subject to sufficiently complete market oversight and discipline. (New Zealand has recently implemented this approach.) Government could reasonably withdraw from the oversight process only if market forces could promptly and accurately identify bank risks and asset values, including the extent to which one bank's problems implied difficulties for other financial institutions. I return to these issues in section 8, after discussing the available evidence about the efficacy of market discipline.

I conclude this discussion of bank corporate governance with a caution. Banking firms are subject to an unusual combination of supervisory systems. In evaluating potential changes to this arrangement, we cannot permit the "best" to become enemy of the "good." No supervisory system will be perfect. Any system operated by human beings will make mistakes, failing to identify behavioral changes promptly, failing to price risks accurately, etc. Our choice is not between one perfect system and one imperfect one, but rather between two imperfect systems. The question is: which system produces better results a larger proportion of the time? or which system yields better results for the most important type of problems?

2. SOURCES OF MARKET INFORMATION ABOUT BANK CONDITION

For banks with publicly traded securities, stock and bond prices (returns) provide the most obvious sources of market information about the firm's condition.¹⁷ The

16. At an IMF Conference in early 1997, Federal Reserve Board Governor Kelley opined that banks remain quite special in their susceptibility to runs and in the severe consequences that a large-scale banking panic would involve today. Balancing the need for a banking "safety net" to defuse potential bank runs with the need to create the right incentives for banks in assessing and assuming risk is one of the most difficult challenges we face as central bankers (Kelley 1997).

17. It is interesting to note that FDICIA mandated that regulators track equity market values: "Each appropriate Federal banking agency shall, for all insured depository institutions and depository institution

stock holdings of *institutional investors* might also reflect changes in informed opinions about bank quality, again for banking firms with traded equity. However, market assessments can also be inferred for banks without deeply traded equity or liabilities. The stock holdings of *bank insiders* should reflect their own assessment of the bank's condition, and this information could be obtained for all banks.¹⁸ *Credit-rating agencies* (Moody's, Standard and Poor's, etc. for large investors and services like Veribanc for retail customers) rate the CDs, debentures, and commercial paper programs of banks and their holding companies. They also rate individual banks, even when the banks have no intention of selling negotiable CDs in public markets.¹⁹ The *average rate* paid on large (uninsured) deposits provides a market assessment of bank condition even when those deposits do not trade publicly.²⁰ Finally, some *loan customers* should be sensitive to their bank's financial condition. Loan commitments and letters of credit are worth more from a sound bank, and established customer relationships will be more valuable when the bank seems likely to endure for a long time.²¹

Skeptics about the role of market forces in bank supervision often point out that market prices are unavailable for most U.S. banking firms. While most U.S. banking companies have no publicly traded securities, the majority of banking system assets belong to banks whose parents trade in the market.²² Table 2 reports the extent of traded banks for the third quarter of 1995. Only about 15 percent of all banks are associated with holding companies whose equity shares trade on the NYSE, AMEX, or NASDAQ. However, these 1,599 banks hold more than three trillion dollars in assets, comprising fully three-quarters of the U.S. banking system.

Table 3 reports similar information about the number and proportion of banking firms with publicly traded debt outstanding. Although these banks are few in number (only 1.08 percent of all banking firms in 1995), they hold more than half of all banking system assets.

The fact that market data are not available for many banks does not justify ignoring this information for the largest banking firms, which pose the greatest challenge to federal regulators' supervisory abilities. The next two sections evaluate whether market valuations of these traded securities have been accurate.

holding companies, prescribe . . . to the extent feasible, a minimum ratio of market value to book value for publicly traded shares of the institution or company" (FDICIA Section 39, paragraph b). This requirement was subsequently revoked, along with a number of other "micro-management" efforts embodied in the original legislation.

18. Jordan (1997) concludes that bank insiders had a more accurate assessment of their bank's survival probability than was reflected in market prices, during the real estate-related banking crisis in New England in 1989-90.

19. For example, banks need a rating to issue certain types of letters of credit.

20. In evaluating retail deposit rates, however, we must distinguish between a higher default risk premium and a higher return designed to attract customers from outside the institution's natural market area.

21. Several studies have evaluated the effects of market discipline on a bank's off-balance-sheet commitments (Hassan 1994; Hassan, Karels, and Peterson 1994; Koppenhaver and Stover 1994; Petersen and Rajan 1994; Slovin, Sushka, and Polonchek 1993 present evidence that customer relationships are valuable.

22. The mistaken impression that few banks have traded equity may reflect the rapid expansion in traded banks over the past two decades. Whereas only 157 banks were listed on the NYSE, AMEX, or NASD exchanges in 1975, this number had grown to 378 by 1985, and to 566 by 1995. Most of this growth (87.5 percent) reflects additional NASD listings.

TABLE 2
EQUITY MARKET DATA AVAILABILITY FOR U.S. BANKING FIRMS

Bank's Primary Regulator	Number of Banks	Percentage of All U.S. Banks	Bank Assets	Percentage of U.S. Bank Assets
OCC	639	6.2%	\$1,866 billion	45.1%
Federal Reserve	246	2.4%	\$828 billion	20.0%
FDIC	590	5.7%	\$336 billion	8.1%
Other (e.g. Edge)	124	1.2%	\$69 billion	1.7%
Total	1599	15.4%	\$3,099 billion	75.0%

NOTES: Sample of 344 bank holding companies with traded equity shares as of September 30, 1995 [data derived from the sample in Flannery, Kwan, and Nimalendran (1977)].

3. THE ACCURACY OF MARKET PRICES: LIABILITY RATES AND QUANTITIES

Market discipline in the liability markets may affect either the cost or availability of new funds. Furthermore, the market response to a change in bank risk can occur gradually and simultaneously with changes in risk, or as a sudden withdrawal of deposits after the risk has already reached some critical threshold. Assuming that the typical bank's condition changes slowly, "good" market discipline involves a gradual increase in liability rates, perhaps combined with a steady decline in funds availability. Regulatory discomfort with the effects of market discipline often reflects the assessment that depositors are more likely to withdraw their funds suddenly, in a "panic" to avoid imminent (and irreversible) credit losses. Liability-market discipline will be most effective if investors promptly identify, and react to, default risk changes.

A. *Market Rates on Large Certificates of Deposit*

A number of researchers have investigated whether the interest rates paid on large, uninsured CD obligations of U.S. banks reflect their apparent default risk. These papers shared a basic methodology: cross-sectional regressions of large CD rates on bank accounting data.²³ While the conclusions are not unanimous, most researchers found that measured bank risk affects large CD rates in a plausible fashion (Baer and Brewer 1986; James 1988, 1990; Hannan and Hanweck 1988; Cargill 1989; and Keeley 1990). However, these cross-sectional regression models may share measurement or specification errors, which led Ellis and Flannery (1992) to devise an alternative method for detecting default risk premia. For a sample of six large, money-center banks (with presumptively strong conjectural guarantees), they estimated time series models for daily CD risk premia as a function of innovations in the bank's stock return over the period 1982–1988. Their motivation was straightforward: if equity and CD investors have access to the same information set, news that affects share prices should also have implications for the bank's default probability, and hence should simultaneously affect CD rates. Ellis and Flannery reported results consistent with the

23. Baer and Brewer (1986) and James (1988, 1990) substituted a market measure of risk (the standard deviation of recent equity returns) for some accounting ratios, and found a significant positive effect of equity volatility on CD rates.

TABLE 3
DEBENTURE MARKET DATA AVAILABILITY FOR U.S. BANKING FIRMS

	1985	1995
Total U.S. Banking Assets	\$2460.3	\$4157.2
Banking Assets of Firms with Traded Debentures Outstanding	\$1409.3	\$2267.6
Proportion with Traded Debentures	57.3%	54.5%
Total U.S. Banking Firms	11,089	7,602
Number of Firms with Traded Debentures Outstanding	72	82
Proportion with Traded Debentures	0.65%	1.08%

SOURCE: Lehman Bros. Fixed Income Database, Berger, Kashyap and Scalise (1995).

hypothesis that bank CD rates immediately reflect the information impounded in bank stock prices. Even for TBTF banks, a reduction in the asset portfolio’s value significantly raises CD risk premia.

Since these empirical results apply to the TBTF period of strong conjectural guarantees, they seem to provide strong evidence that creditors value bank default risks.

B. Market Rates on Bank Holding Company Debentures

In light of the evidence that bank CD rates reflect the issuer’s risk, it seemed very likely that bank holding company debentures—which enjoyed a less-secure conjectural guarantee than bank CDs did—would also exhibit similar risk sensitivities. It was therefore quite disturbing when the early studies of bank debentures detected *no relation* between risk premia and other measures of default risk. For example, Avery, Belton, and Goldberg (1988) examined the default risk spreads for about one hundred U.S. bank holding companies’ debentures at year-end 1983 and 1984, and concluded that “the potential for market discipline to effectively and constructively augment regulatory controls via the subordinated note and debenture capital market is *weak* (p. 609, emphasis added). Gorton and Santomero’s (1990) subsequent application of a different (nonlinear) methodology to the same data reached the same conclusion: that there was “little support for the presence of market discipline in the subordinated debt market” (p. 127).

Flannery and Sorescu (1996) (FS) investigated whether the inability to detect significant market disciplinary effects in 1983–4 debenture rates reflected the conjectured protection of the federal regulators’ TBTF policy.²⁴ Numerous thrift and bank failures during the mid-1980s had forced regulators to trim the extent of this conjectural protection by late in the decade (Flannery and Sorescu 1996, pp. 1352–53). To test whether market discipline appeared after investors became exposed to de facto credit risk, FS assembled a data set of debenture risk premia for a cross-section of banks at the end of each year between 1983 and 1991. Both the magnitude of risk premia and their cross-sectional dispersion rose sharply after 1989 (see their Figures 1 and 2).

Regressing debenture spreads on accounting and market measures of bank risk

24. As noted above, this policy was formally announced by the Comptroller in September 1984, but indications that large banks were unlikely to expose creditors to default losses were available earlier, in the cases of Franklin National Bank (1974) and First Pennsylvania (1980).

yielded virtually no explanatory power in the early years. However, once TBTF debenture guarantees receded in 1988–9, private investors began to differentiate clearly among individual banks' creditworthiness. For the post-TBTF years of 1989–91, FS' regressions explain a high proportion of the cross-sectional variation in *SPREAD*. They conclude that market investors *can* evaluate individual banks' credit qualities, but will do so only when they feel that their invested principal is at risk. Put another way, government supervision and guarantees severely compromised private incentives to impound appropriate premia into risky bank debt.

C. Retail Customers' Behavior

Although many observers contend that individual depositors cannot make informed judgments about bank risks, some evidence suggests that this is not entirely correct. Kane (1987) evaluated depositor withdrawals from Ohio thrift institutions during the state's insurance fund crisis of March 1985. He found that depositors "ran" only against state-insured thrifts, whose deposits were rendered less safe when the Ohio Deposit Guarantee Fund failed. Federally insured institutions, and even several *uninsured* thrifts, experienced no unusual withdrawals during the period. Even in crisis situations, therefore, small depositors have been able to discriminate between solvent and insolvent depository institutions.

A similar conclusion emerges from Sommer's (1996) study of the rates charged for retail property-liability insurance coverage. Sommer observed that the value of this insurance varies directly with the insurer's probability of surviving until a loss might occur.²⁵ As an empirical matter, he found that companies with higher leverage and/or more portfolio risk charge lower insurance premia, consistent with the hypothesis that "market discipline exists in insurance markets that is likely to restrain insurer risk taking" (p. 512).

Two studies evaluated the availability of large retail deposits to thrift institutions during the 1980s' crisis.²⁶ Goldberg and Hudgins (1996) examined the proportion of *uninsured* deposits at failed thrifts during the two- to four-year period preceding their failure. For 261 S&Ls that failed (or received some type of federal assistance) between January 1988 and March 1990, mean uninsured deposits fell from 6.73 percent of total deposits sixteen quarters before failure to 2.13 percent of total deposits at the quarter-end preceding failure. The authors also estimated a multivariate model to explain thrift reliance on uninsured deposits, and found that an institution's (estimated) probability of failure had a significantly negative effect on uninsured deposit reliance in all sample years (1986–1989). This study indicates that nominally uninsured depositors can identify failing thrifts up to four years before their actual failure.²⁷

25. Although most states provide some type of guaranty funds to cover the obligations of failed insurance companies, Sommer (1996, pp. 503–4) observes that these do not substitute perfectly for the survival of one's insurance company.

26. I discuss these two papers here, rather than in connection with the large-CD pricing studies in section A, because they do not evaluate CD rates. Moreover, the samples in these studies have probably gathered information on large retail CDs, rather than true wholesale deposits.

27. Contrast this finding with Helfer's (1996) admission that "Of the 1,617 banks that failed in 1980 through 1994, 36 percent had CAMEL ratings of '1' or '2' two years prior to failure." Cole and Gunther (1998) similarly conclude that CAMEL ratings become "stale" after six months.

Cook and Spellman (1996) evaluated the rates offered by S&Ls on six- to twelve-month CDs with a minimum principal of \$80,000. (Note that at least some of these instruments were fully insured under the \$100,000-per-account limit.) For the twenty months (January 1987–August 1988) during which the FSLIC became insolvent and was recapitalized, Cook and Spellman found that the S&Ls' offering rates generally rose and fell together, as if depositors believed that all thrifts' default probabilities varied in a similar way over time. However, nonlinear regression analysis revealed that this shared variation in CD rates resulted from changes in the credibility of the FSLIC deposit insurance, *not* in the default premia required from individual issuers. Indeed, the estimated default probability for similar groups of S&Ls remained quite stable over the period, suggesting that depositors were pricing the *compound* probability of an S&L's default and the FSLIC's insolvency.

4. THE ACCURACY OF MARKET PRICES: BANK EQUITY RETURNS

Special bank regulations are sometimes motivated by the unusual “informational opacity” of bank assets, which can make bank portfolios difficult for outside investors to value.²⁸ When (if) a bank's fixed claimants perceive that asset values have fallen, they will rationally “run” to redeem their claims before the bank is closed. Government supervision may be required if one bank's problems lead depositors mistakenly to infer that other banks are similarly affected. Historically, these concerns have been most commonly expressed in terms of bank depositors, whose insurance protection has limited their incentives to investigate their bank's condition and to run when they perceive insolvency. However, bank equity-holders—who claim the most junior elements of a bank's opaque cash flow—are affected by the same inference problems. One bank's failure might influence the market's assessment of other banks for either of two reasons (Aharoney and Swary 1983). First, if other banks share a risk exposure with the failed bank, their claimants would *rationally* perceive lower values. This constitutes a normal and desirable aspect of efficient financial markets, in which investors' actions convey appropriate signals to bank managers. Second, other banks' claimants might *mistakenly* infer that their own banks' quality had deteriorated. Subsequent deposit withdrawals or higher required deposit rates could impose (inappropriate) costs on other banks, perhaps inducing their failure.²⁹

Examining stockholders' responses to information events can therefore indicate how “uninsured” outsiders assess bank values. Two questions are particularly important:

1. Do market investors *accurately* assess the implications of new information about a bank's (opaque) asset portfolio?
2. Is bad news contagious across banks? In other words, does news about one bank cause investors to make *inappropriate inferences* about the condition of other banks?

28. Note that information asymmetry need not cause outsiders to misestimate a bank's total asset value. Even large uncertainties about the value of *individual* loans may correspond to slight uncertainty (asymmetry) about the overall *portfolio's* value if the individual valuation errors are not too highly correlated.

29. Kaufman (1994) extensively reviews the literature on contagious bank runs.

This section next discusses the question of rational inference in the context of equity market studies from outside the financial sector. Section 3C describes bank-related equity studies, and the final part of this section evaluates the implications of these studies for the efficacy of private market discipline.

The implications of stock market event studies can be carried over to depositor behavior with two potential caveats, both of which apply primarily to very thinly capitalized institutions. First, stock and bond values need not move in the same direction when a bank's condition changes. If the bank has a higher probability of failure, further increases in risk can raise equity value while it depresses deposit value. Second, equity holders "own" a put option (written by FDIC) whose value is impounded in stock price. New information about FDIC policies toward failing banks may change the value of this put option, which simultaneously affects many banks' stock prices. Note, however, that this common movement in share values does not imply irrational (unfounded) "contagion"!

A. General Studies of Intra-Industry Information Flows

Developments at one firm carry informational implications for other firms exposed to similar effects. Szewczyk (1992) investigates the impact of security issuance announcements on the stock returns of *nonissuers* in the same industry. It is well established that securities issuance elicits a negative abnormal return to the issuer's equity, which could imply either good or bad news for other firms in the same industry. If a firm's security issuance reflects a cash shortage resulting from poor operating results, competitors might expect to benefit from the issuer's weakness. Conversely, if the issuer's need for external cash reflects a general decline in the value of assets throughout the industry, competing firms' stock prices should fall because the issuer's action conveys new information about the *industry's* overall condition. Based on his analysis of 484 public security offerings, Szewczyk concludes that

the abnormal returns of the announcing firms and the industry portfolios [of nonissuers] are positively correlated. This suggests that investors draw inferences about the general prospects of the announcing firm and its industry competitors. (p. 1945)

In other words, firms in the same industry share sensitivities to the same factors, and the actions or fortunes of one firm imply revaluations for the others.³⁰

A similar conclusion emerges from Firth's (1996) evaluation of intra-industry reactions when one firm changes its dividend. Using a set of 649 dividend changes over the period 1980–1991 (543 dividend increases and 106 decreases), Firth measures the impact on the announcing firm's stock price and on the stock prices of nonannouncers in the same industry. He concludes that dividend changes significantly affect the announcing firm's stock value. They also affect nonannouncers in the same direction, though to a smaller extent. A nonannouncers' abnormal return more closely resembles the announcer's, the greater is the historical correlation between the two firms' equity values (p. 210). Firth concludes

30. Lang and Stulz (1992) analyze cross-firm effects of bankruptcy announcements.

that dividend changes not only signal information about the future cash flows to the firm concerned but they also signal valuation changes for other companies in the same industry. These spillover effects help explain the comovement of security prices within industry segments. (p. 190)

Szewczyk (1992) and Firth (1996) document the principle that news about one firm will promptly influence similar firms' market values. This feature of efficient markets means that cross-sectional correlation in bank equity returns does not necessarily imply irrational contagion.

B. Contagion Studies from Other Industries

Market reactions to announcements in industries other than banking provide a good backdrop against which to evaluate studies of bank equity returns and potential contagion. Most of these studies conclude that investors make rational judgments about the implications of one firm's experience for the value of other, similar firms in the same industry.

Bowen, Castanias, and Daley (1983) evaluate the impact of the nuclear accident at General Public Utilities' Three Mile Island (TMI) generating plant on March 28, 1979. While their sample of all electric utilities showed a modest negative return following the TMI accident, the effects were most severe among two subsamples: the set of utilities that relied heavily on nuclear generating capacity, and the set of utilities that were involved with nuclear plants constructed by the same contractor as TMI. Bowen, Castanias, and Daley conclude that "[T]he evidence is consistent with TMI having caused a significant, downward price shift related to a utility's commitment to nuclear power and/or BW [the contractor] plants" (p. 106).

A similarly discriminating response is documented by Asness and Smirlock's (1991) evaluation of the real estate investment trust (REIT) industry. The Residential Resources Mortgage Investment Corporation was a highly leveraged REIT holding primarily residual tranches of mortgage CMOs. "Res Res" failed in January 1989, only seven months after its founding. Asness and Smirlock construct a portfolio of thirty-five other REITs, for which they measure a (statistically insignificant) abnormal return of -1.42 percent on the date of Residential Resources' failure. When the REITs were separated into three groups, the only group with a significant negative return was the one most closely resembling Residential Resources. In particular, the portfolio of eighteen unleveraged equity REITs had an insignificant announcement day return of -0.32 percent, eight leveraged REITs which did *not* hold CMO residuals had an insignificant -1.20 percent return, and nine "residual" REITs resembling the failed firm had an abnormal return of -3.82 percent, significant at the 1 percent level. Cross-sectional regressions of the individual REITs' announcement returns show a significant negative effect of REIT leverage on stock return, consistent with the hypothesis that Residential Resources' failure conveyed more information about more similar REITs.

Fenn and Cole (1994) provide information that is particularly relevant to small depositor behavior when they evaluate the impact of two life insurance companies' asset

quality problems on other firms in the industry. In January 1990, First Executive announced its decision to write down the book value of its junk bond portfolio by more than \$500 million (2.8 percent of the company's total assets). Nine months later, Travelers reserved \$650 million (1.8 percent of its total assets) for expected losses on its commercial real estate portfolio. These events generated large negative returns to the announcing companies' stock prices: within a few days, Executive's stock price had declined by half and Travelers' stock price had fallen by a third. Stock declines of this magnitude were hard to reconcile with the size of the announced asset quality problems alone, particularly since the market had prior indications that the junk bond and commercial real estate sectors were in trouble. Fenn and Cole hypothesized that investors expected life insurance customers would react to the announced writedowns by reducing their demand for insurance savings products. They argued that one particular class of customer—purchasers of guaranteed investment contracts (GICs)—would be most likely to react to asset quality problems, since GICs were basically a savings product which competed directly with (federally insured) bank deposits and a broad range of other market alternatives. The authors called this their “policyholder-response hypothesis.”

When Fenn and Cole evaluated the cross-sectional variation in insurance companies' stock responses to the First Executive and Travelers announcements, they found that the other companies' stock price reactions reflected both their asset and liability features. When the news concerned junk bond valuations, insurers holding higher proportions of junk bonds were harder hit, and when the news concerned commercial real estate the firms with high real estate holdings were most affected. In both cases, however, the insurers' stock prices were affected by the *interaction* of problem assets with the level of its GIC obligations. A higher concentration of problem assets reduced other life insurers' market value, but the reduction was particularly severe at firms which also had substantial GIC balances outstanding. Fenn and Cole concluded that stock market investors rationally re-valued life insurers on the basis of new information about their assets and on the conjectured response of policyholders to that new information. These were not “contagious” revaluations of all insurers, but rather a measured response consistent with informationally efficient expectations revision.

The potentially contagious impact of Hurricane Andrew (August 1992) on property-liability insurance companies has been evaluated in two papers—Lamb (1995) and Angbazo and Narayanan (1996)—with conflicting results. Both papers identified insurance companies with premiums written in the two states affected by Andrew (Florida and Louisiana), and compared these firms' equity value changes to those of insurers with no losses from the storm. Lamb (1995) evaluated thirty-seven traded property-liability companies, of which twenty had direct premiums written in Florida or Louisiana. He found significant negative returns only for the directly affected subsample. Moreover, insurers with more business in the affected states suffered more negative abnormal returns. Lamb concluded that “The market efficiently interpreted the information generated by the hurricane and discriminated among property-liability insurers based on the existence and magnitude of insurance written” (p. 120).

Angbazo and Narayanan (1996) assemble a sample of forty-five traded property-

liability insurers, of which twenty-three had written coverage in Florida or Louisiana. While the “exposed” group of insurers suffered a more negative average return ($AR = -0.71\%$, $p < 0.01$) than the unexposed group ($AR = -0.54\%$, $p < 0.01$), the two groups’ abnormal returns were statistically indistinguishable.³¹ Angbazo and Narayanan interpret the significant negative response of unexposed insurers to Andrew’s damage as reflecting “industry-wide *contagion* effects since [Andrew] significantly affected most insurers, regardless of whether these firms had any claims exposure in the hurricane-affected states” (p. 619, emphasis in the original).

C. *Equity Market Studies of Bank Contagion*³²

With the preceding two sections as background, I now survey three types of bank equity study, emphasizing their implications for outsiders’ ability to value bank assets accurately. These studies fall naturally into three areas: reactions of stock prices to announcements about loan loss provisions, to new information about the value of LDC loans, and to the failure of large banking firms.

1. *Loan Loss Provisions.* If bank assets are informationally opaque, periodic announcements of loan loss provision should convey new information to market analysts about the loan portfolio’s value. Docking, Hirschey, and Jones (1997) examined 578 bank loan loss reserve announcements from the period 1985–1990.³³ They found that one bank’s surprising addition to its loan loss reserves (LLR) induced a negative effect on *non*-announcing banks’ stock prices. (That is, higher-than-expected reserve provisions cause nonannouncing banks’ stock prices to fall.) But this effect is absent among money center banks. They conjecture that the

relatively modest negative information transfer effects associated with money-center bank LLR announcements may reflect the influence of intense security analyst coverage and the continual flow of money-center bank information (p. 238)

By contrast, regional bank LLR announcements generated significantly negative returns at nonannouncing regional banks, and these effects were stronger for banks located closer to the announcing bank. Docking, Hirschey, and Jones thus concluded that the spillover effects associated with loan loss reserve announcements reflected investors’ rational revisions of estimated loan values, and not general contagion.

2. *LDC Loans.* Researchers have studied the equity market’s reactions to two major LDC debt payment moratoria: Mexico in 1982 and Brazil in 1987. Analyses of the Mexican moratorium’s effect on U.S. bank stock prices yielded conflicting results. Cornell and Shapiro (1986) found no immediate effect of a bank’s *Latin American* loan exposure on its stock. (They did conclude, however, that market valuations reflected cross-sectional differences in Latin American exposures after about one year.) Smirlock and Kaufold (1987) measured each bank’s holding of *Mexican* loans, and

31. In their cross-sectional regression, Angbazo and Narayanan found no significant effect of a company’s loss exposure on its abnormal return (Table 3), though it is puzzling why they exclude their *EXP*(osure) variable from the “exposed” firm regression.

32. Other studies related to the efficiency of bank stock prices include those by Fraser and Richards (1985), Pettway (1976, 1980), Pettway and Sinky (1980), and Simons and Cross (1991).

33. Related papers include Grammatikos and Saunders (1990), and Madura and McDaniels (1989).

conclude that the cross-sectional distribution of stock returns on the moratorium announcement day accurately reflected these exposures. Finally, Bruner and Simms (1987) found that equity market prices did not fully reflect Mexican loan exposures until about six days after the moratorium announcement. They conclude that "These results suggest that markets learn rapidly. In the absence of reliable information on loan exposure, investors appear to respond to surprising news rationally and quickly" (p. 54).

Following Mexico's default, the SEC mandated that banks include in their 10-Q and 10-K reports detailed information about their exposure to overseas loans. When Brazil declared its debt moratorium on February 20, 1987, therefore, investors could readily determine their banks' relevant loan holdings. Musumeci and Sinkey (1990) provide strong evidence that equity investors promptly and accurately impounded each bank's credit exposure into its stock price. For their sample of the largest twenty-five U.S. bank holding companies, the estimated announcement returns were proportional to reported bank exposures. [Karafiath, Mynatt, and Smith (1991) replicate the equity market findings for a larger sample of forty-six banks, using a more sophisticated statistical model.] Furthermore, Musumeci and Sinkey gathered market prices for Brazilian debt in the first three months of 1987, and find that changes in bank equity prices between January and March of 1987 were correlated with changes in the market value of the bank's Brazilian debt holdings.

3. Bank Failures. If bank investors are prone to contagious inferences, we might expect this tendency to be most apparent when a prominent financial firm fails. Aharoney and Swary (1983) analyzed bank stock reactions to three large failures from the 1970s. In two instances (the U.S. National Bank of San Diego in 1973 and Hamilton Bancshares in 1976), they found other banks' stock prices did not respond at all. The third case (Franklin National Bank in 1974) resulted from foreign exchange trading losses, which could also have affected other banks. Aharoney and Swary concluded that "the observed drop in solvent bank stock prices may be interpreted as investors' reaction to a common type of unfavorable signal, rather than a pure contagion effect" (p. 321).

Swary (1986) employed a similar methodology to evaluate market reactions to Continental Illinois' 1984 troubles. Other banks experienced negative weekly stock returns when Continental's problems were announced, and these negative returns were larger for poorly capitalized banks. He interpreted this cross-sectional pattern of negative market reactions as rational conjectures about the banks' true asset values, rather than shareholders' fear of depositor runs. Wall and Petersen (1990) evaluated *daily* returns during the week in which Continental failed, and concluded that most of the other banks' abnormal returns during that week were associated with developments in the Latin American loan market. They found no "contagion" after controlling for these other announcements. Instead, Wall and Peterson showed that banks with large reliance on purchased funds had *positive* abnormal returns when the full guarantee of all Continental creditors was announced. Because large depositors had suffered credit losses in Penn Square's 1982 failure, Continental's treatment seemed to imply a re-extension of conjectural guarantees.

Aharoney and Swary (1996) provided further evidence that bank equity investors make rational inferences from announced banking problems. They investigated the potential contagion effects associated with five large bank failures in the southwestern United States between 1986 and 1989.³⁴ The authors identified thirty-three traded bank holding companies headquartered in the FDIC's southwest region and computed the abnormal equity return to each surviving bank on each of ten "critical event" dates associated with the five failures. They then regressed each critical event's vector of abnormal returns on three characteristics of the surviving bank holding companies: the physical distance between the BHC's headquarters and that of the troubled institution; the BHC's total assets; and the BHC's book equity ratio. Aharoney and Swary found that bank-specific characteristics often explained part of the cross-sectional variation in surviving banks' abnormal returns (see their Table 2), and concluded that investors were making rational inferences about the value of surviving banks.

Two other papers similarly concluded that investors rationally incorporate one bank's failure into their valuation of other banks. Investigating the 1988 failure of First RepublicBank of Texas, Dickinson, Peterson, and Christiansen (1991) found that only Texas and southwestern regional banks experienced negative equity returns when First RepublicBank's problems initially surfaced, presumably because investors viewed those problems as implying negative effects for banks with similar loan portfolios. Karafiath and Glascock (1989) assessed the impact of Penn Square's 1982 failure and find significant negative effects only for the bank's upstream loan correspondents and a portfolio of Texas banks holding similar energy-related loans.³⁵

D. Assessment of Equity Studies

The literature provides broad, though not unanimous, support for the hypothesis that bank equity holders respond rationally to announced news. Market revaluations occur rapidly, and exhibit cross-sectional patterns consistent with rational inferences about the implications of announcements for other firms. These results seem to reject the possibility that bank investors (and, by extension, sophisticated depositors) routinely engage in "pure contagion" inferences about all banks, which augurs well for the efficacy of market discipline. Nevertheless, an event study cannot determine that investors' revised assessments of the banks are *correct*, but only that they *seem to be reasonable*. Perhaps these studies have only identified the misinformation on which investors base their mistaken inferences! Documenting that the nonannouncers' price changes do not systematically reverse themselves would support the rationality of these initial revisions, but we have no such evidence. Equity event studies do contain one positive implication for market discipline: investors apparently incorporate rele-

34. The failed banks were the First National Bank and Trust Company of Oklahoma City, and four Texas banks: First National Bank of Midland, First City Bankcorporation, First Republic Bankcorporation, and MCorp.

35. Another interesting equity market study investigated the impact of the problem bank list's disclosure in January 1976. Murphy (1979) matches twenty-four banks which were revealed to be on the Comptroller of the Currency's "problem bank list" with a similar-sized nonproblem bank in the same geographic area. He finds that the "problem" banks' stock prices fell but the control group's did not, consistent with the hypothesis that investors did not make contagious inferences about the nonproblem banks.

vant information promptly into stock prices. Disclosure requirements and publicizing supervisory information about banking firms' opaque assets may therefore help investors form even more accurate assessments of bank values (DeYoung et al. 1998).

5. U.S. HISTORICAL EVIDENCE ABOUT MARKET EFFICACY

When discussing bank solvency and the need for a government safety net we must recognize that "banking" services can be provided in a variety of institutional forms. Institutional arrangements endogenously reflect government policies. Do banks operate with high leverage for basic economic reasons, or simply because high leverage permits them to capture greater subsidies from the government safety net? If government deposit insurance were eliminated, would banks provide AAA-rated subsidiaries to issue deposits to unsophisticated customers? And so forth. Information about how market institutions behave under alternative regulatory regimes must come from past U.S. arrangements, or we must look abroad. In the interests of space, I consider only the domestic, historical evidence.³⁶

Prior to the Banking Act of 1933, national bank shareholders were exposed to double liability, in contrast to the limited liability enjoyed today by virtually all corporate shareholders. Many states also permitted banks to organize themselves with extended shareholder liability.³⁷ Esty (1997) analyzed the impact of bank shareholders' liability on risk taking and portfolio choice during the period 1900–1915. He concluded that more severe shareholder liability rules induced bankers to select less-risky portfolios and to operate with less leverage, thereby reducing the bank's default probability. The absence of payoffs to a "go for broke" strategy as a bank's capital approached zero sometimes lead national banks to close themselves voluntarily, in order to avoid further shareholder losses.

Wilson and Kane (1997) evaluated the economic factors generating the political pressure to abandon double liability for national bank shareholders. Before the 1930s, depositors and bank owners had voluntarily negotiated extended liability in a joint effort to minimize the costs and distortions of bankruptcy. Wilson and Kane contended that the growth of individual banks (which necessitated a larger number of shareholders) and the dramatic deterioration in economic conditions combined to reduce the economic value of double liability. Depositors became less likely to collect from heterogeneous, dispersed shareholders, and solvent equity holders found it more costly to monitor management and their co-owners. These effects were particularly important for large banks, and Congress' elimination of extended liability in 1934 reflected its economic obsolescence.

Saunders and Wilson (1996) evaluated deposit flows at individual banks during the last U.S. occurrence of major bank runs, 1929–1933. For each bank that failed during this period, the authors chose a bank headquartered in the same geographic area that

36. Some relevant foreign experiences are described in Miller (1996) and Jao (1974).

37. Extended and double shareholder liability required shareholders to make additional cash contributions to a firm that became insolvent.

survived past 1933. Saunders and Wilson reported that the banks that eventually failed had lower deposit growth up to three years before their demise, suggesting that depositors responded to perceived bank weaknesses in advance of the bank's failure date. Calomiris and Mason (1997) analyzed a June 1932 incident during which twenty-six of Chicago's commercial banks failed during a single week. The authors labeled this incident a "panic" because substantial deposits were withdrawn from both sound and unsound banks. However, Calomiris and Mason contended that the banks that eventually failed were truly insolvent, while the survivors were not. In addition, objective disparities between the failures and the survivors existed before the crisis struck: the failed banks had lower equity prices (Tobin's q) as much as eighteen months before the panic, and their total interest costs during the second half of 1931 significantly exceeded those for surviving banks.

6. THE TIMELINESS OF REGULATORY VERSUS MARKET ASSESSMENTS OF BANK CONDITION

Under the present system of joint public-private supervision, both government and market agents have incentives and opportunities to collect information about various dimensions of bank performance. Since the agents' goals and incentives differ, it would be surprising if they collected just the same information, at just the same times. Bond investors seem to have the most in common with supervisors, since both groups are concerned primarily with the bank's probability of failure. Equity market investors, by contrast, receive their payoffs in the bank's *non*-default states and may have only limited interest in the type of analysis undertaken by government supervisors (and vice versa).

Hard information on the relative timeliness of private versus government information about bank quality is difficult to obtain, partly because the regulatory agencies have generally been unwilling to share their quality assessments with academic researchers.³⁸ Two early studies provided little evidence that the examiner information in CAMEL ratings was reflected in market assessments of bank condition. For fifteen large banks, Hirschhorn (1987) found that CAMELs were contemporaneously correlated with stock returns but provided no predictive power. Cargill (1989) sought to explain bank CD rates using financial ratios and CAMEL ratings, and found that the CAMELs added no explanatory power to the regression. In contrast to these early studies, a few recent papers indicate that market and regulatory assessments complement one another quite well. Supervisors may obtain information (particularly through on-site examinations) that is not available to market analysts, and the question arises how quickly that information works its way into market prices. Similarly, market analysts may be more forward-looking than examiners, making market assessments better predictors of changes in bank condition.

38. Regulators have been reluctant to disclose current CAMEL ratings, for example, because they fear that publication of any negative assessments could become self-fulfilling prophecies, causing the affected banks to suffer even greater difficulties. Though the publication of current CAMEL ratings may raise difficult questions, providing historical ratings to researchers under a pledge of confidentiality seems less problematic.

Berger and Davies (1994) examined the behavior of BHC stock prices in the eight-week period following a federal examination of the holding company's lead bank, and concluded that examiners were particularly good at discovering information that managers would prefer to conceal. For a sample of 390 bank examinations during the period 1985–1989, Berger and Davies found that a firm's stock price significantly declined in the wake of a CAMEL downgrade, but showed no abnormal return following a CAMEL upgrade. While the market subsequently learned the “bad” information underlying an examiner's downgrade, managers apparently released “good” information promptly to the market. At least for a little while, therefore, examiners have a richer information set than the market does. Dahl, Hanweck, and O'Keefe (1995) detected a similar asymmetry: large allocations to a bank's loan loss allowance tend to occur only in the wake of an on-site examination. They concluded that examiners exert additional discipline, beyond that provided by a bank's private auditors.³⁹ Flannery and Houston (1998) provided further evidence that market investors view bank financial statements as more informative when examiners have recently visited the bank.

Cole and Gunther (1998) compared two models for predicting bank failure. One model uses only Call Report accounting information, while the other relies on CAMEL ratings. They found that recent CAMEL ratings could better predict bank failures than published financial information could. After about six months, however, CAMEL ratings become “stale” and add no information to what is contained in financial statements alone. While a system of frequent on-site examinations would therefore generate valuable nonpublic information, long intervals between exams tends to make the average de facto value of supervisory information fall short of its theoretical value.

DeYoung et al. (1998) (DFLS) provided further evidence that examiners are most likely to detect new “bad” information, and that examination information becomes stale in a relatively short time. Using quarterly data for national banks over the period 1989–1995, they regressed each assigned CAMEL rating on financial data obtained from the Call Report immediately preceding the underlying exam. The regression residuals were then used to measure examiners' “private information.” Aggregating this private information across a holding company's national bank subsidiaries produced a measure of the examiners' *average* private information, which was then used to explain the holding company's debenture risk premium. DFLS demonstrated that examiners' private information significantly affects contemporaneous *and subsequent* changes in debenture spreads. Moreover, the impact of private information diminishes with the average time since examination. This study concluded that on-site examinations provide an informational advantage to government supervisors, under the current institutional arrangements.⁴⁰

A similar study by Berger, Davies, and Flannery (1998) (BDF) compared BOPEC

39. An alternative interpretation, which underlies many of the studies discussed in this section, is that examiners inappropriately force large loan loss recognitions. Bank managers are more likely to convince their auditors that a loan problem will resolve itself.

40. A further question is whether market agents could effectively replace this private information if government were to curtail their supervision of large banking firms.

ratings of bank holding company quality with stock and bond market assessments, over the period 1989:Q4 through 1992:Q2. For a sample of 184 large bank holding companies, they gathered four market assessments of bank holding company condition: abnormal stock return, proportion of common equity shares held by insiders, proportion of common equity shares held by institutional (13-F) investors, and the Moody's rating on outstanding debentures. These data were matched with BOPEC ratings, the dates of on-site holding company inspections, and financial (Y-9) information. BDF's Granger causality tests indicated that bond market ratings and BOPEC ratings each reflected information that was not fully incorporated into the other. Optimal forecasts of changes in BHC condition would therefore be based on both market and supervisory information sets, suggesting that bond investors and supervisors are concerned with similar aspects of BHC performance. Importantly, these results indicate that bond market assessments *do not* systematically lag assessments derived from on-site inspections of the holding company! BDF's findings about the relative timeliness of equity market versus supervisory assessments of bank condition were quite different. Neither equity market data nor supervisory assessments Granger cause the other, consistent with the hypothesis that equity investors care primarily about a BHC's nondefault states, while supervisors are concerned primarily with the default states. BDF also found that bond and stock returns significantly predict subsequent changes in key holding company performance variables, while the typical BOPEC rating does not.⁴¹

Taken all together, the recent research reviewed in this section suggests that market assessments have at least a plausible chance of providing timely, accurate information that supplements the supervisory agencies' traditional ways of gathering and assessing bank quality.

7. THE RELATIVE EFFECTIVENESS OF MARKET VERSUS GOVERNMENTAL DISCIPLINE

The evidence thus far strongly suggests that market assessments of bank condition are accurate and timely, and the potential for irrationally contagious runs seems very limited. Regulators should therefore incorporate market information more centrally into their supervisory assessments of firm condition. However, effective corporate governance (discipline) requires not only the recognition of a firm's condition, but also reliable mechanisms by which to reverse or control undesirable changes in firm condition.⁴² While we generally assume that market forces can effectively discipline nonfinancial firms, few studies have addressed this issue for banking firms. Are the problems more severe in this sector? Is there reason to think that a higher (different) standard is appropriate for banking versus nonbanking firms?

Crabbe and Post (1994) illustrated the effectiveness of market discipline in a study which examined the funding responses of large bank holding companies to commer-

41. As in Cole and Gunther (1998) and DeYoung, Flannery, Lang, and Sorescu (1998), the age of BOPEC ratings matters. When BDF restrict their sample to BHC with very recent on-site inspections, BOPEC ratings predict future performance more accurately than the market measures do.

42. I thank Ben Esty for emphasizing the importance of this topic.

cial paper program downgrades between 1984 and 1991. They found that a rating downgrade caused the holding company's stock of outstanding commercial paper to fall sharply (17.74 percent over the ensuing quarter), and that this fall was financed by a reduction in booked assets. In short, firms whose credit quality deteriorates were quickly forced to contract their balance sheets.

Market discipline can also be provided by other firms operating in the same industry. Gorton and Mullineaux (1987) discussed the case of nineteenth-century bank clearinghouses, which monitored their members and provided liquidity in response to deposit outflows. A similar arrangement governs life insurance companies operating with state guarantee funds. In some states, the losses to life insurance guarantee funds will be borne entirely by the surviving companies, while in other states the taxpayers essentially bear any fund losses. If industry members effectively monitor and discipline one another when they have an incentive to do so, we should find that insurance companies operating in states with taxpayer-underwritten guarantee funds operate with higher risk. Brewer, Mondschean, and Strahan (1997) studied this issue and conclude that the guarantee fund's structure significantly affects life insurance company risk taking.

Swindle (1995) provided some evidence that supervisors can effect desired changes in a bank's condition, by demonstrating that banks with poor Capital components in their CAMEL ratings tended to raise equity in subsequent quarters. She also found that "capital regulation at national banks was less effective than at state-chartered banks" (pp. 137–38), indicating that supervisors exercise discretion in implementing regulations.

Research that explicitly compares the efficacy of market versus government discipline is rare. One recent example is Billett, Garfinkel, and O'Neal (1998), who examined the impact of bond rating changes on a bank holding company's funding composition. They find that downgraded banks shifted out of large, uninsured deposits and increased their issuance of insured, retail balances during the quarter of the downgrade and the following one. Upgraded banks made the converse change: they reduced insured deposit obligations and increased large, uninsured deposit balances. This behavior implies that bankers viewed regulatory supervision (discipline) as less costly than market supervision. The authors point out that market discipline cannot substantially affect bank behavior if bankers can readily shift into less-disciplinary (that is, government-regulated) funding sources when problems arise.

As I shall emphasize below, the scarcity of empirical evidence about the *relative* efficacy of market versus supervisory discipline makes it very difficult to argue for greater reliance on one versus the other type of corporate governance. In this area, much further research is needed.

8. INCORPORATING MARKET INFORMATION INTO PRUDENTIAL SUPERVISION

The studies reviewed thus far generally support the following conclusions.

1. Investors understand changes in bank condition and promptly impound their implications into equity and liability prices.

2. Market investors do not seem to make irrationally contagious inferences about banking firms.
3. On-site examinations seem to generate some new information beyond what public investors know, but this advantage is not very large.

How, then, might market information be incorporated into the bank supervisory process?

A radical reform of U.S. financial supervision would have the federal agencies withdraw entirely from the supervision and insurance of banking firms which were judged to have sufficient market oversight.⁴³ Credibly withdrawing safety net protections would provide investors with the incentive to control banking firms, and the empirical evidence indicates that they clearly have the ability to do so. Banking firms' corporate governance structures would become less homogeneous, as banks and market investors negotiated freely over capital adequacy, internal risk control procedures, corporate structure, and permissible activities. Regulators should be unconcerned by such innovations *provided* that they can credibly withdraw all implicit guarantees.

However, continuing concerns about systemic risk and ex post political pressure for supervisors to "do something" about major financial failures will make it rational for investors to incorporate some type of conjectured government support for large financial firms into their corporate governance structures.⁴⁴ Government cannot, as a practical matter, withdraw completely from prudential supervision because it cannot credibly shed all contingent obligations to the banking sector. The relevant policy question is how to use market information to complement existing procedures for monitoring and controlling large banking firms.

Figure 1 describes the process of identifying a problem bank and producing a supervisory response. At date 0 the bank initiates policies which will eventually place it in danger of failing. While this event passes unnoticed at the time, regulatory oversight seeks to identify such changes in bank condition or risk exposure as soon as possible. Suppose a system of on-site government examinations would identify the bank's increased failure probability at the point B. The supervisory system's *Recognition Lag* is therefore $[0, B]$. Anecdotal evidence strongly suggests that supervisors cannot usually impose corrective action at B, but must wait until subsequent developments have made it obvious that the bank has a real problem (O'Keefe and Reidhill 1997). Call this interval $[B, D]$ the *Action Lag*.

Systematically combining market and supervisory information could shorten the Recognition Lag and provide more accurate forecasts of future changes in bank condition.⁴⁵ (In terms of Figure 1, the expanded information set would shorten the Recognition Lag to $[0, A]$.) While regulators currently employ market information in their

43. This approach yields a two-tier system of banks, each subject to a different type of corporate governance. While identifying which banks have "sufficient market oversight" poses difficult practical issues, the important policy point is that most banking *assets* could be subjected to expanded market discipline.

44. My skepticism about narrow banks as a solution to the systemic risk problem derives from my assessment that regulators will intervene when large financial firms become troubled, out of a (misguided?) fear of systemic contagion. Penati and Protapadakis (1988) present a model in which conjectural guarantees cause bank loans to be underpriced.

45. One can readily show that sampling a larger number of noisy signals of bank condition often provides a more accurate estimate of the bank's true value.

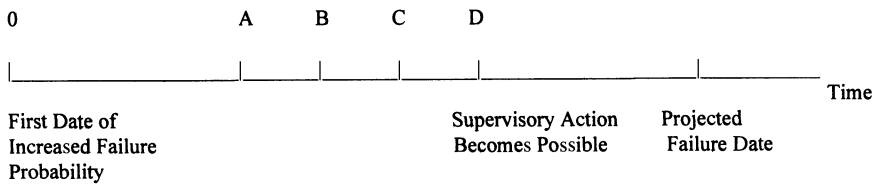


FIG. 1. Monitoring and Failure Prevention

off-site supervisory assessments for a small number of large institutions, these data are neither formally nor prominently incorporated into the supervisory process. Despite many supervisory officials’ skepticism about the quality of market information about bank condition, the studies reviewed here find that

- (a) Retail depositors can reliably identify troubled institutions several years before failure (Kane 1987; Saunders and Wilson 1996; Calomiris and Mason 1997).
- (b) Abnormal stock returns, insider holdings, and institutional holdings predict bank performance several quarters in advance (Berger, Davies, and Flannery 1998).
- (c) Banks substitute insured, retail deposits for uninsured funding after their bond ratings have declined (Billett, Garfinkel, and O’Neal 1998).

Market assessments can also be used to shorten the Action Lag in the supervisory process. O’Keefe and Reidhill (1997) contend that supervisors usually know about problem situations long before they can effect solutions. While many of these delays are associated with the U.S. legal system, critics also assert that regulators are reluctant to recognize problems that have occurred “on their watch.” Broad agreement that market opinions are generally accurate would frequently constrain regulators’ discretion in cases where the market has identified substantial problems.⁴⁶ Supervisors interested in acting quickly could appeal to market information to justify the appropriateness of their concerns, shortening the typical Action Lag from [B, D] in Figure 1 to [B, C].⁴⁷

A third implication of this paper’s analysis is that government regulators should consider whether they can help market analysts come to more accurate or timely assessments of banks’ condition. Could market analysts more readily evaluate and compare financial firms if they all reported according to the same type of market value accounting? What about setting other types of disclosure standards? Would the prompt publication of CAMEL or BOPEC ratings provide the market with useful information? These are difficult questions, about which we currently have no real information.

This particular lack of information notwithstanding, the available evidence pro-

46. Though regulators frequently resist efforts to limit their discretion (which is why, for example, many opposed Wall’s (1989) suggested introduction of puttable subordinate debentures), a precommitment to implement policies without discretion can be optimal.

47. The Action Lag may reflect administrative delays within the agency, or examiners’ legal inability to impose costly oversight on a bank while its accounting condition remains “good.” Market evidence that investors also view the bank as troubled could increase the pressure for administrators to act quickly, and provide justification that corrective measures are needed.

vides little support for the notion that market forces understand and control banking firms less well than they do nonbanking firms. Nor do supervisors generally understand large banks' true condition more clearly than market investors do. Bank investors' behavior suggests that governments can transfer a greater proportion of default risks to equity and liability holders without inducing financial system instability. Although transferring additional bank credit risks to market investors entails some difficult implementation issues, one point is clear: banking firms' owners and large creditors must feel that they are at risk. This must be implemented via continued depositor payouts (even for large banks!), while inflation continues to erode the real value of the FDIC coverage limit.⁴⁸ Additional reliance on subordinated debentures for bank capital would also increase the impact of market discipline, since regulatory procedures already expose these investors to credit losses. Moreover, bank debenture holders are natural allies of government supervisors, since they share similar risk exposures.

Further research can guide a continuing revision to supervisory policies. The following list defines some of the topics requiring more detailed investigation.

- (1) Government regulation should respond to identified market failures. What type of firms (or activities) require government oversight to supplement the usual market forms of corporate governance? In particular, are banking assets more informationally opaque than other firms? What are the causes and scope of systemic risk?
- (2) How much market oversight is required before supervisors can safely reduce their own supervision of a banking firm?
- (3) We need much more information about the relative timeliness and accuracy of market versus supervisory assessments of bank condition.
- (4) What is the *effectiveness* of market discipline in influencing bank risk-taking? How does that effectiveness compare with government supervisors' ability to implement corrective actions promptly?
- (5) At a conceptual level, how should policymakers compare two imperfect systems of oversight? How should they identify and compare the errors that each type of system is likely to make?
- (6) Could supervisors enhance private assessments of bank condition by releasing some of their (currently confidential) information? Would other types of mandatory disclosures improve private market discipline?
- (7) Would supervisors' more explicit use of market information adversely influence private incentives to generate that information? In other words, does some type of "Lucas critique" apply to the use of private information by government agents to discipline private firms? Bernanke and Woodford (1997) analyze a related question in the context of monetary policy.
- (8) The finance literature has evaluated how bankruptcy rules affect the bargaining game between the owners and creditors of a distressed firm. Banking firms operate under their own set of bankruptcy rules. What do these rules imply about the efficacy of private versus public supervision?

48. Today's \$100,000 limit was introduced in 1980. The current (1997) value of \$100,000 equals 66,659 1980 dollars.

- (9) What is the foreign evidence about the efficacy of market analysts' efforts to identify and control banking firms' risk exposures?

9. SUMMARY

This paper has reviewed the literature on the ability of market forces to identify and control risk taking in banking firms. The evidence shows that bank share prices generally behave very similarly to the equity prices of nonbanks: they adjust promptly to new information, and make rational inferences about the implications of that information for related firms. Bank liability investors also behave rationally. Even in the presence of conjectural federal guarantees, large CD rates sensibly reflect bank risks. Bank debenture rates came to reflect bank condition once conjectural guarantees were lifted from these instruments. Even small, retail depositors have been shown to behave rationally when some intermediaries exhibit solvency problems. There is little evidence that broadly contagious "runs" would be a problem for banks even in the absence of the federal safety net protection for depositors.

Market assessments of bank condition are generally rational, and exhibit a timeliness that compares well with supervisory assessments. Bond ratings are particularly complementary to supervisory assessments of bank condition, and equity market variables provide relatively good predictions of bank holding company performance. Given the limited incentives under which private analysts have operated to date, it seems plausible that they would do an even better job if they were more subject to default losses. This is not, however, a one-way street. Greater access to regulatory information may also improve market analysts' assessments. Some evidence indicates that the market values examiner-related financial statements more highly than those produced by private auditors, and subordinated debenture rates tend to lag behind examiners' private assessments of bank condition.

The oversight of banking firms could be improved if federal supervisors systematically incorporated more market information into their analyses and action plans. While this trend has begun in recent years (at least in the United States), regulators should move further toward formally combining private with public oversight at larger banking firms. The financial market's information flows seem sufficient to warrant greater reliance on private analysts' judgments, particularly if regulators continue to curtail de facto guarantees of individual financial firms. The next logical step in designing an optimal public/private oversight structure is to commission some targeted research, which can confirm the conclusions reviewed here and determine their practical implications.

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